BioOne RESEARCH EVOLVED

Endophilic *Anopheles* Mosquitoes in Guinea Bissau, West Africa, in Relation to Human Housing Conditions

Author(s): Katinka Pålsson, Thomas G. T. Jaenson, Francisco Dias, Ane T. Laugen, and Anders Björkman

Source: Journal of Medical Entomology, 41(4):746-752.
Published By: Entomological Society of America
DOI: http://dx.doi.org/10.1603/0022-2585-41.4.746
URL: http://www.bioone.org/doi/full/10.1603/0022-2585-41.4.746

BioOne (www.bioone.org) is a nonprofit, online aggregation of core research in the biological, ecological, and environmental sciences. BioOne provides a sustainable online platform for over 170 journals and books published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Web site, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/page/terms_of_use.

Usage of BioOne content is strictly limited to personal, educational, and non-commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Endophilic *Anopheles* Mosquitoes in Guinea Bissau, West Africa, in Relation to Human Housing Conditions

KATINKA PÅLSSON,[1] THOMAS G.T. JAENSON,[1, 2] FRANCISCO DIAS,[3] ANE T. LAUGEN,[4] AND ANDERS BJÖRKMAN[5]

J. Med. Entomol. 41(4): 746–752 (2004)

ABSTRACT Environmental risk factors associated with increased malaria mosquito (*Anopheles*) abundance indoors were studied in a suburban area, Antula, of Guinea Bissau, during the rainy seasons of 1993–1995. All bedrooms in 30 houses were searched for resting mosquitoes three times each year. The most abundant mosquito was *An. gambiae s.s.* Significantly greater numbers of resting mosquitoes were present in rooms with open eaves and in houses with a well on the compound. Pigs were the most common domestic animals in Antula. Presence of pigs in a house was associated with increased mosquito abundance in the bedrooms of the same house. The abundance of female mosquitoes also increased with increasing human biomass per square meter of bedroom area.

KEY WORDS *Anopheles*, domestic animals, entomological risk factors, Guinea Bissau, malaria

THE PREVALENCE OF HUMAN malaria can differ significantly between closely adjacent localities. This phenomenon has been recorded in several studies, e.g., in Liberia (Björkman et al. 1985), Papua New Guinea (Cattani et al. 1986), and The Gambia (Greenwood et al. 1987). One reason for such differences may be different human behavior or cultural affiliations, which will influence the individuals' exposure to infective mosquitoes (Trape et al. 1992, Lindsay et al. 1995).

In addition to human behavior, ecological factors such as the presence or absence of domestic animals may influence abundance of mosquitoes and thus human inoculation rates. Studies from Pakistan (Hewitt et al. 1994, Bouma et al. 1995) and The Gambia (Adiamah et al. 1993) have shown that domestic animals in the house or compound were associated with increased malaria parasite prevalence and incidence of clinical malaria episodes. Bøgh et al. (2001) found no zooprophylactic effect by cattle on malaria vectors in The Gambia. Adiamah et al. (1993) found that children who had experienced malaria tended to live more often in mud-walled buildings and in bedrooms without ceilings. Gamage-Mendis et al. (1991) recorded lower mosquito densities and malaria incidence rates in the inhabitants of large affluent houses with brick and plaster walls. Lindsay et al. (1995) and Smith et al. (1995) have mentioned several environmental factors that explain differences in mosquito densities between houses. In The Gambia, increased exposure to malaria mosquitoes during the rainy season was associated with living near a mosquito breeding site, living in a house with open eaves or without a ceiling, the absence of wood smoke indoors, and presence of a storeroom adjacent to the bedroom (Lindsay et al. 1995). In Tanzania, areas of high malaria mosquito abundance tended to be in the low-lying areas with persistent breeding sites (Smith et al. 1995).

The aim of this study was to investigate how indoor mosquito abundance, and thereby human exposure to infective malaria mosquitoes, may be influenced by different local environmental factors, particularly house construction, number of people sleeping in a room, presence of domestic animals, and house location. This work was carried out in a suburban area, Antula, of Bissau, the capital of Guinea Bissau. Previous studies involving local environmental factors potentially affecting the malaria risk have generally been performed within 1 yr or have been designed as a case (malaria)/control study. Here we investigate a suburban area during three rainy seasons on 3 consecutive yr. This permitted us to observe any changes in mosquito abundance between the years.

Materials and Methods

Study Area. The study was carried out in a suburban area (Antula; 11°50′ N, 15°30′ W), 5 km north of Bissau, Guinea Bissau, West Africa. The study area is coastal and surrounded by flat lowlands affected by the daily tides. On the eastern side, Antula is bordered by a large rice field, mangrove swamps, and the Canal do Imperial, which flows into the Rio Geba. On the western

[1] Department of Systematic Zoology, Evolutionary Biology Centre, Uppsala University, Norbyvägen 18 D, SE-752 36 Uppsala, Sweden.
[2] E-mail: Thomas.Jaenson@ebc.uu.se.
[3] Laboratório Nacional de Saúde Pública (LNSP), Bissau, Guinea Bissau.
[4] Department of Population Biology, Evolutionary Biology Centre, Uppsala University, Norbyvägen 18 D, SE-752 36 Uppsala, Sweden.
[5] Infectious Disease Unit, Department of Medicine, Karolinska Hospital, Stockholm, Sweden.

0022-2585/04/0746–0752$04.00/0 © 2004 Entomological Society of America

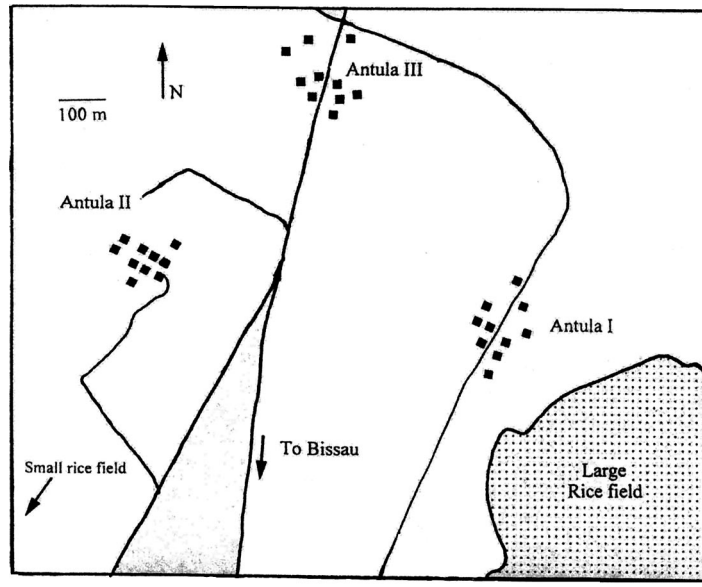


Fig. 1. Location of the three groups of houses (I–III) in the study area, Antula, Guinea Bissau.

side, there is a small rice field. The population of Antula represents several different ethnic groups. Cultivated plots of land are normally located close to each house, with manioc, millet, maize, and groundnuts as the predominate crops. Domestic animals were present in all three areas and included pigs, goats, ducks, chicken, cats, and dogs. Cattle were only kept in one house in Antula. Antula is divided into three subareas, i.e., Antula-Bono-Somec, Antula-Bono, and Antula-Bono-Cancala (Antula I–III, respectively), which are ≈650–850 m apart (Fig. 1). In each subarea, 10 houses were selected for the study: Antula I, houses 1–10; Antula II, houses 11–20; Antula III, houses 21–30 (Fig. 1). Houses 10, 13, 14, and 29 were subdivided into two households; all the other houses harbored one household each.

Both anthropological factors and characteristics of houses and surrounding areas were recorded to determine entomological risk factors for mosquito abundance. All inhabitants in the study households were registered with name, age, sex, house and room number, and ethnic affiliation. Children <16 yr old represented 52% of the whole population and 57, 52, and 47% in areas I–III, respectively.

The use of anti-mosquito measures was also recorded. Unimpregnated mosquito nets of varying quality, but usually of poor condition, were used in all houses by most of the people (94%), except in one house in area III in 1993, where nobody used nets. A total of 4% used other measures, e.g., anti-mosquito coils, mosquito repelling leaves, or wood smoke.

In July 1994, we replaced 158 new unimpregnated nets around the beds in Antula I–III. In October 1994, a house-building project started in Antula II and III; after removal of a lot of vegetation and large trees, three new roads were constructed. Antula I was indirectly affected by the building project because traffic with heavy vehicles increased in the area.

Every house was characterized with respect to construction including building material, type of roof and floor, number and types of rooms, windows and doors open or closed at night, and presence or absence of ceiling and open eaves. The area of each room was measured to the closest 0.5 m^2. Presence and species of domestic animals inside the house at night were recorded. The effect of potential mosquito breeding sites was recorded as distance to rice fields and presence of a well on the compound. In Antula I, 6 of 11 households had wells for drinking water on their own compounds, whereas only 4 of 23 households had their own wells in Antula II and III. This information was used to determine entomological risk factors for mosquito abundance. The study was performed during the rainy seasons of 1993–1995. The rainy season usually lasts from early June to late October or early November (Fig. 2).

Mosquito Collections. Resting mosquitoes were collected indoors with battery-operated aspirators and torches in the mornings (0700–1000 hours) in all rooms where people had been sleeping the previous night. Collections were performed three times each year. Collections in 1993 were as follows: first, 28 September–8 October; second, 9–19 October; third, 20–29 October. Collections in 1994 were as follows: first, 19 August–4 September; second, 6–21 October; third, 7–18 November. Collections in 1995 were as follows: first, 25 September–11 October; second, 12–25 October; third, 30 October–10 November. The number and ages of people who had been sleeping in each room during the previous night were recorded, and total human biomass per room was estimated by weight.

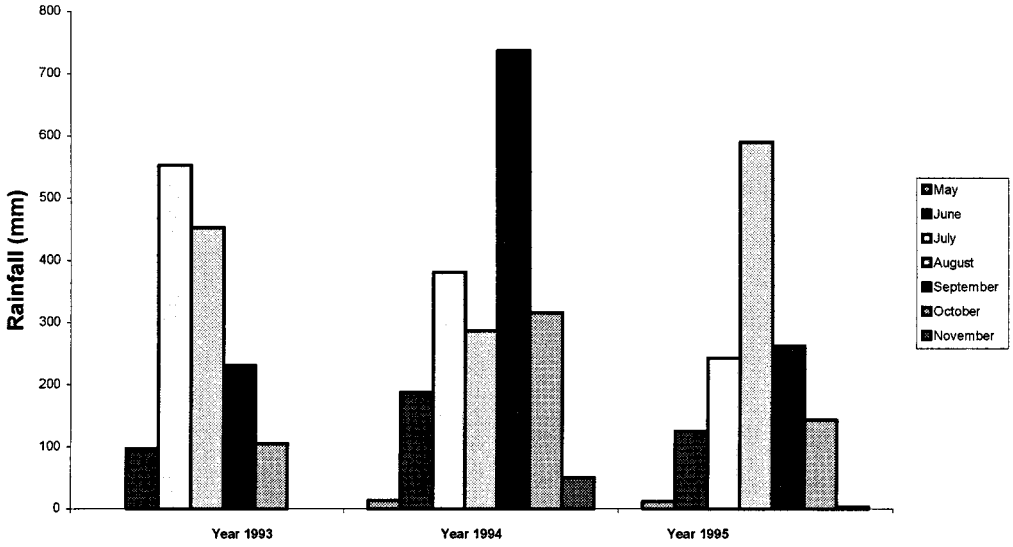


Fig. 2. Precipitation (mm) per month (May–November 1993–1995) at Bissau Meteorological Station.

Crepuscular and nocturnal mosquitoes were also collected outdoors. First, UV light traps (Miniature Black Light Traps; JW Hock Co., Gainesville, FL) were operating from 1900 to 0700 hours. Light trap collections were carried out during eight trap-nights (traps × nights) in 1993, six in 1994, and two in 1995. Second, anthropophilic mosquitoes landing on human baits were captured outdoors between 1815 and 0810 hours, during the nights of 13 September and 21 October 1994. Two persons, acting as both bait and collector, were sitting 1–2 m apart and ≈1 m outside house 12. Torches and battery-operated aspirators were used to collect mosquitoes that landed on the skin or clothes of the collectors.

Mosquitoes were killed with chloroform vapor, put into 1.5-ml vials, and stored on silica gel at −20°C. A total of 2,009 females collected in 1993 and 1994 were identified by microscopy (Coluzzi 1964, Bryan 1980). Of the *An. gambiae s.l.* identified, a total of 147 were also identified by polymerase chain reaction (PCR) (Paskewitz and Collins 1990, Scott et al. 1993). In 1995, a total of 354 randomly selected *An. gambiae s.l.* mosquitoes were identified by PCR (Pålsson et al. 1998).

Statistical Analyses. To analyze the potential risk factors for exposure to *Anopheles* mosquitoes, the number of female mosquitoes sampled in each room at each sampling event was used as response variable in a type III generalized linear model with a log link function and a Poisson error structure as implemented in the GENMOD procedure in the SAS statistical package (version 6.12; SAS Institute 1996). Initially, we ran a model including year and month of sampling as well as area, house, and room nested within house as factors. There were no differences between areas, and this factor was excluded from further analyses. Because there were significant differences between houses (df = 25, χ^2 = 91.92, P = 0.0001) and rooms within houses (df = 76, χ^2 = 357.42, P = 0.0001), various characterizations of the houses and rooms were included in a larger model. The distance from each house to the two rice fields were fitted as continuous variables (covariates). The presence of wells on the compound and open eaves, windows, and doors were all fitted as categorical variables. Furthermore, material of floor (four levels) and mattresses (three levels) and the presence of pigs and goats were also included as categorical variables. Finally, human biomass per square meter was fitted as a covariate.

Results

Mosquito Abundance and Species Composition. Totals of 5,676, 1,543, and 3,338 mosquitoes were collected indoors by aspirator in 1993, 1994, and 1995, respectively. Most (78.7–98.1%) were female *Anopheles* (Table 1). The mean values for the different collection occasions in the three areas are shown in Fig. 3.

Table 1. Total number and percentage of mosquitoes (Culicidae) collected indoors during August–November in 1993, 1994, and 1995 at Antula, Guinea Bissau

	Female *Anopheles*	Male *Anopheles*	*Culex* spp.
1993			
Antula I	2,037 (94.4%)	12 (0.6%)	1
Antula II	2,286 (97.3%)	29 (1.2%)	35 (1.5%)
Antula III	1,246 (97.7%)	9 (0.7%)	21 (1.6%)
Total	5,569 (98.1%)	50 (0.9%)	57 (1.0%)
1994			
Antula I	696 (95.5%)	12 (1.6%)	21 (2.9%)
Antula II	358 (94.2%)	10 (2.6%)	12 (3.2%)
Antula III	382 (88.0%)	4 (0.9%)	48 (11.1%)
Total	1,436 (93.1%)	26 (1.7%)	81 (5.2%)
1995			
Antula I	1,124 (94.1%)	11 (0.9%)	60 (5.0%)
Antula II	611 (86.9%)	81 (11.5%)	11 (1.6%)
Antula III	1,133 (78.7%)	262 (18.2%)	45 (3.1%)
Total	2,868 (85.9%)	354 (10.6%)	116 (3.5%)

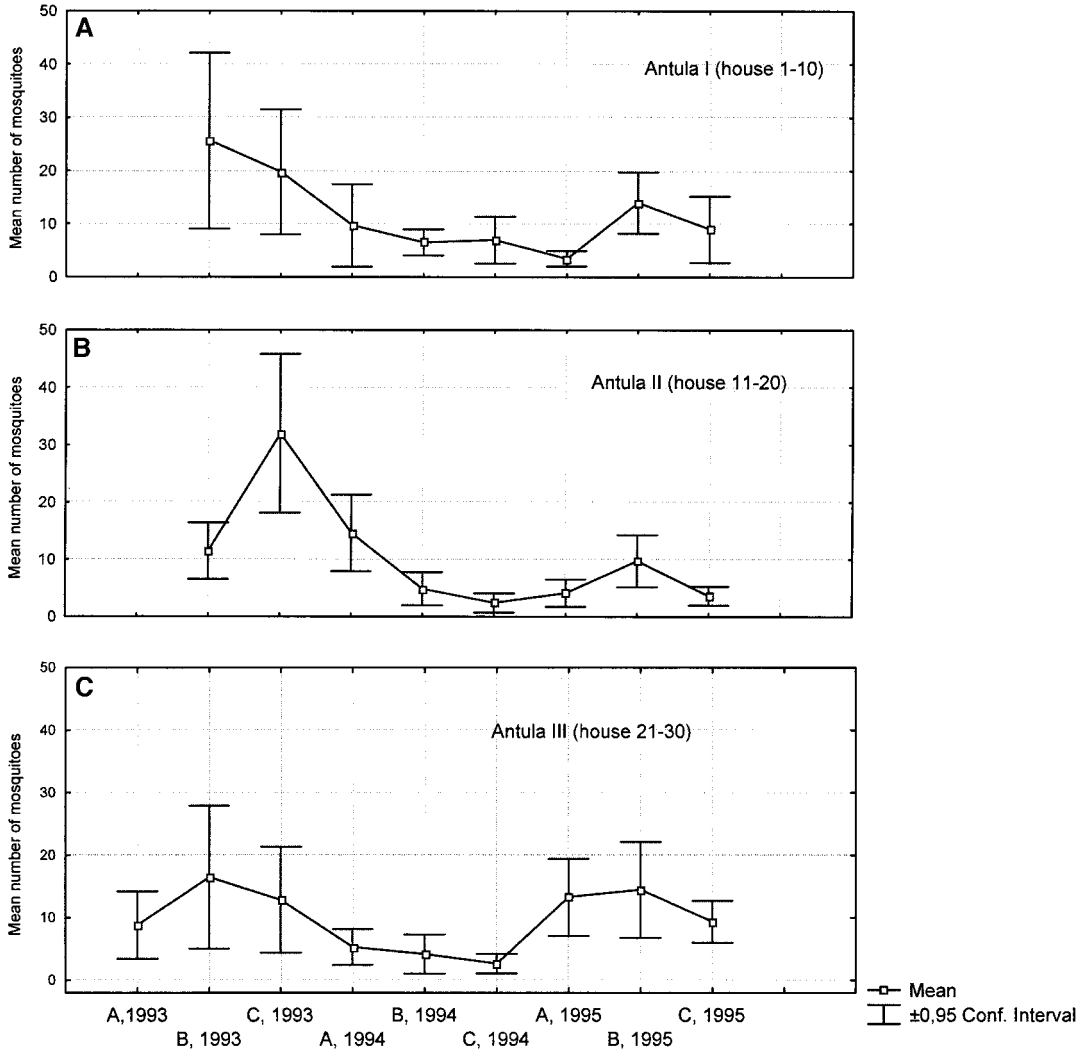


Fig. 3. Mean number of mosquitoes collected at Antula, Guinea Bissau, A, B, and C denote collection occasions.

The predominant culicid taxa collected, by UV light traps outdoors, were *An. gambiae s.l.* and *An. squamosus* (Table 2).

Mosquitoes attracted to and landing on human bait were collected during the nights of 13–14 September and 21–22 October 1994, when 279 and 26 mosquitoes, respectively were captured. The taxa identified were *An. gambiae* (63%), other *Anopheles* spp. (1%), *Mansonia* spp. (25%), *Aedes* spp. (6%), and *Culex* spp. (7%; Fig. 4). The outdoor human bait catches showed that *Anopheles* spp. had high activity throughout the night (22:15–06:10; Fig. 4).

Table 2. Outdoor light trap collections during the rainy seasons of 1993–1994

		An. gambiae s.l.	*An. squamosus*	*An. ziemanni*	*An. coustani*	*An. rufipes*	*An. smithii*	*An. pharoensis*	*Culex* spp. f	Unid. female mosquitoes	Total mosquito females	Male mosquitoes
1993	*N*	1455	335	77	5	12	1	3	726	74	2678	206
(8 nights)	% of tot.						—					
	females	54.3	12.5	2.9	0.2	0.4		0.1	27.1	2.8		
1994	*N*	71	119	69	1	—	—		215	59	534	9
(6 nights)	% of tot.						—				—	—
	females	13.3	22.3	12.9	0.2				40.3	11.0		

Number and percentage of female *Anopheles* spp., female *Culex* spp., male mosquitoes, unidentified mosquitoes and total female mosquitoes. *N*, no. of mosquitoes.

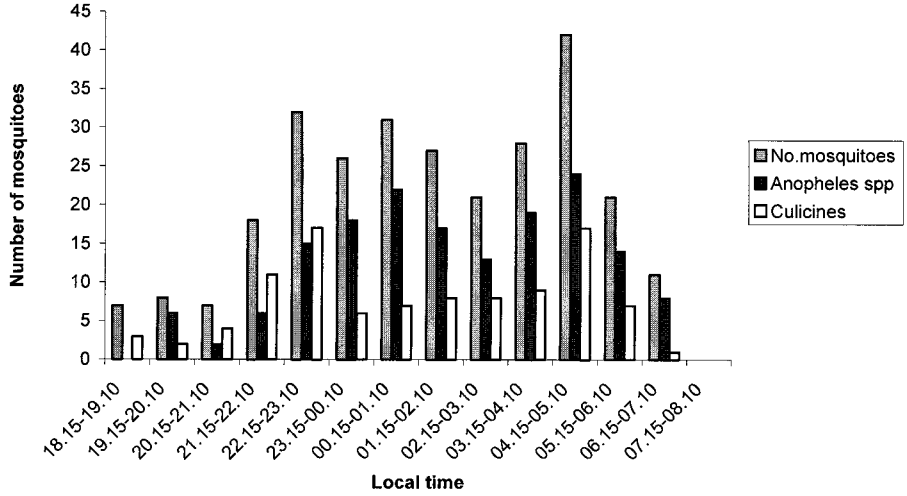


Fig. 4. Mosquito diel activity at Antula, Guinea Bissau, 13–14 September 1994.

The abundance of *Anopheles* in bedrooms varied during the 3 yr and with month within years (Table 3). In 1994, significantly fewer mosquitoes were caught than in the other 2 yr.

Species were identified by microscopy. Among the identified *An. gambiae s.l.*, 147 specimens were identified by PCR. Among these, 94.6% were *An. gambiae s.s.* and 5.4% were *An. melas.* Eighteen *An. squamosus* Theobald, 14 *An. pharoensis* Theobald, 2 *An. coustani* Laveran, 2 *An. ziemanni* Grünberg, 1 *An. funestus* Giles, and 1 *An. rufipes* Gough were the other *Anopheles* species among 2,009 females collected and identified in 1993 and 1994. In 1995, *An. gambiae s.s* was the most prevalent (98.0%) species, followed by *An.melas* (2.0%), among 354 anophelines identified by PCR (Pålsson et al. 1998).

Entomological Risk Factors. Mosquito counts varied both between years and between months within years (Table 3). The mean ± SE distance between houses and the large rice field and the small rice field was 549.7 ± 9.0 m (range, 120–990 m) and 1092.7 ± 5.9 m (range, 820–1,420 m), respectively. There was no effect of distance to rice fields, whereas the presence of a well on the compound significantly increased mosquito numbers indoors (Table 3). Furthermore, there were significantly more mosquitoes in bedrooms with open eaves (Table 3), but there was no interaction effect between presence of a well and open eaves (data not shown). Building material of floors did not affect mosquito abundance, and the same was the case for mattress material (Table 3). The presence of pigs significantly increased the number of mosquitoes collected in bedrooms (Table 3), whereas goats did not have any effect. The abundance of female mosquitoes also increased with increasing human biomass per square meter (Table 3).

Table 3. Generalized linear model (type III) of the number of female *Anopheles* mosquitoes caught in bedrooms in 30 houses in a suburban area in Guinea-Bissau in 1993, 1994, and 1995

	df	χ^2	*P*
Year	2	63.60	0.0001
Month	3	16.82	0.0008
Distance to large rice field	1	2.44	0.1181
Distance to small rice field	1	0.80	0.3705
Well on compound	1	23.12	0.0001
Open eaves	1	48.63	0.0001
Floor material	3	3.53	0.3161
Mat material	2	2.62	0.2704
Presence of pigs	1	17.63	0.0001
Presence of goats	1	1.08	0.2991
Human biomass/m^2	1	63.60	0.0001

Discussion

Great differences in mosquito abundance were observed among the different houses in all three areas. Great variations among rooms within a house were also recorded. Increased exposure to mosquitoes was associated with sleeping in rooms with open eaves. Similar results have been reported from The Gambia (Lindsay et al. 1995, Lindsay and Snow 1988). Small pools of water are often present around wells and are potential breeding sites for *Anopheles* mosquitoes. Having a well on the compound was associated with increased mosquito abundance. Pigs were the most common domestic animals in Antula. A relation was found between mosquito abundance and presence of pigs in a house. In contrast, the presence of goats in a house was not associated with increased mosquito numbers, possibly because the number of goats indoors was relatively small. Normally the pigs, goats, and chickens were kept inside the main house at night, whereas the cattle were kept in an adjacent shed. In Pakistan, Hewitt et al. (1994) showed that biting rates of *An. stephensi* Liston on people increased when cattle or goats were kept in the vicinity of the house. In the North-West Frontier province of Pakistan, Bouma and Rowland (1995) found that malaria prevalence was significantly higher among families who owned cattle. The zooprophylaxis theory (World Health Or-

ganization 1982) failed, and it seemed that a proportion of the highly zoophilic *An. culicifacies* Giles and *An. stephensi* (Reisen and Boreham 1982), when present in large numbers, were feeding on nearby humans. Only one household in Antula had cattle, but mosquito abundance in that particular house was not exceptionally high. In Antula, the most abundant malaria vector is *An. gambiae s.s.*, which has been shown to be a highly anthropophilic species (Gillies and Coetzee 1987). It can be assumed that most of these mosquitoes, found in the houses with pigs, would feed on humans.

Lindsay et al. (1995) suggested that the relative position of the house to mosquito breeding sites (irrigated rice field) might be a factor significantly affecting mosquito abundance. Trape et al. (1992) showed that the abundance of *An. arabiensis* in Dakar was greater in houses closest to a permanent marsh. However, Adiamah et al. (1993) did not find proximity to the nearest swamp to be a risk factor for malaria. At Antula, distance from a study house to the rice fields was not correlated with *Anopheles* abundance in that particular house. The vast majority of mosquitoes collected indoors in this study belonged to *An. gambiae s.s.* Both *An. gambiae s.s.* and *An. arabiensis* breed in rice fields. However, when the rice is over 1 m tall, it seems to inhibit breeding (Gillies and Coetzee 1987, Klinkenberg et al. 2003). The fact that mosquito counts were performed at the end of the rainy season when the rice was tall may explain why distance from rice field did not affect mosquito abundance in the houses sampled in this study. The prevailing wind direction may also be important (Lindsay et al. 1995). The large rice field close to the study area is to the east of Antula, and the wind during the rainy season comes from the west or southwest. However, breeding sites are not restricted to rice fields or permanent marshland. *An. gambiae s.s.* breeds in small water collections, e.g., borrow-pits, car tracks, and hoof prints. Such breeding sites are virtually impossible to eliminate in the tropics during the rainy season. This could also be one of the reasons why no correlation between the distance to rice field and mosquito abundance in a house was found.

Smaller numbers of mosquitoes were collected indoors in 1994 than in 1993 and 1995 (Table 1). In 1994, the rainy season was atypical, with a lot of rain in September. Variations in mosquito abundance over time exist. Jaenson et al. (1994) recorded lower mosquito abundance indoors in 1991 compared with 1990, despite the higher rainfall in 1991. More mosquitoes were also caught in light traps outdoors in 1993 than in 1994 in Antula. This was probably because of more frequent rain in 1994 on nights when the light traps were operating. In India, Husainy (1978) found that peaks of rain and extremely high or low temperatures reduced the abundance of *An. aconitus* Doenitz. Variation in mosquito abundance was observed at Antula both between years and within years, i.e., among collection occasions. In 1994, a building project started, with a lot of clearing of vegetation, building of new roads in the area, and frequent vehicle traffic. The change of the environment remained in 1995.

In a previous study in Guinea Bissau, Da Cruz Ferreira et al. (1948) recorded that *An. gambiae s.l.* are mainly nocturnal, starting to bite in the late afternoon, continuing the whole night, "until some hours" after sunrise, and that *An. gambiae s.l.* were also biting indoors during daytime. In our study outdoors at Antula, no *An. gambiae s.l.* were observed biting before 1915 or after 0710 hours, although daytime biting by *An.gambiae s.l.* has been observed in other locations in Guinea Bissau (T.G.T. Jaenson, unpublished data). A similar biting pattern of *An. gambiae s.l.* was recorded in The Gambia (Lindsay et al. 1995), where biting peaks occurred at 0130–0230 and 0530 hours, and in Senegal, where Trape et al. (1994) found that peak biting was between 0100 and 0600 hours.

Acknowledgments

We thank R. B. dos Santos (Veterinary Services), M. J. Gomes (LNSP, Bissau), and A. Paulo (Ministry of Public Health, Guinea Bissau) for great help and support; J. Cá (Antula) for excellent assistance in the field; J. Kowalski (Karolinska Institute, Stockholm) and T. H. Ringsby (Norges Teknisk-Naturvitenskaplige Universitet, Trondheim) for advice on statistics, and two anonymous reviewers for constructive comments. This work was supported by grants from Lennander's Stipendiefond and by grants to T.G.T. Jaenson from The Swedish Natural Science Research Council, The Swedish International Development Cooperation Agency, The Swedish Institute for Infectious Disease Control, and The Swedish Research Council for Environment, Agricultural Sciences and Spatial Planning (Formas).

References Cited

Adiamah, J. H., K. A. Koram, M. C. Thomson, S. W. Lindsay, J. Todd, and B. M. Greenwood. 1993. Entomological risk factors for severe malaria in a peri-urban area of The Gambia. Ann. Trop. Med. Parasitol. 87: 491–500.

Björkman, A., P. Hedman, J. Brohult, M. Willcox, I. Diamant, P. O. Pehrson, L. Rombo, and E. Bengtsson. 1985. Different malarial control activities in an area of Liberia—effects on malariometric parameters. Ann. Trop. Med. Parasitol. 79: 239–246.

Bøgh, C., S. E. Clarke, M. Pinder, F. Sanyang, and S. W. Lindsay. 2001. Effect of passive zooprophylaxis on malaria transmission in The Gambia. J. Med. Entomol. 38: 822–828.

Bouma, M., and M. Rowland. 1995. Failure of passive zooprophylaxis: cattle ownership in Pakistan is associated with a higher prevalence of malaria. Trans. Roy. Soc. Trop. Med. Hyg. 89: 351–353.

Bryan, J. H. 1980. Use of the palpal ratio and the number of pale bands on the palps in separating *Anopheles gambiae* Giles *s.s.* and *Anopheles melas* Theobald (Diptera: Culicidae). Mosquito Syst. 12: 155–163.

Cattani, J. A., J. S. Moir, F. D. Gibson, M. Ginny, J. Paino, W. Davidson, and M. P. Alpers. 1986. Small area variations in the epidemiology of malaria in Madang Province. Papua New Guinea Med. J. 29: 11–17.

Coluzzi, M. 1964. Morphological divergences in the *Anopheles gambiae* complex. Riv. Malariol. 43: 197–232.

Da Cruz Ferreira, F. S., A. R. Pinto, and C. Lehmann de Almeida. 1948. Alguns dados sobre a biologia do *Anoph-*

eles gambiae, da cidade de Bissau e arredores em relação com transmissão da malaria e filariase linfatica. Am. Inst. Med. Trop. 5: 223–250.

Gamage-Mendis, A. C., R. Carter, C. Mendis, A.P.K. De Zoysa, P.R.J. Herath, and K. Mendis. 1991. Clustering of malaria infections within an endemic population: risk of malaria associated with the type of housing construction. Am. J. Trop. Med. Hyg. 45: 77–85.

Gillies M. T., and M. Coetzee. 1987. A supplement to the Anophelinae of Africa south of the Sahara. Publications of the South African Institute for Medical Research. No. 55. The South African Institute for Medical Research. Johannesburg, South Africa.

Greenwood, B. M., A. K. Bradley, A. M. Greenwood, P. Byass, K. Jammeh, K. Marsh, S. Tulloch, F. S. Oldfield, and R. Hayes. 1987. Mortality and morbidity from malaria among children in a rural area in The Gambia West Africa. Trans. Roy. Soc. Trop. Med. Hyg. 81: 478–486.

Hewitt, S., M. Kamal, N. Muhammad, and M. Rowland. 1994. An entomological investigation of the likely impact of cattle ownership on malaria in an Afghan refugee camp in the North West Frontier Province of Pakistan. Med. Vet. Entomol. 8: 160–164.

Husainy, Z. H. 1978. Observations on *Anopheles aconitus* (Diptera: Culicidae) in Bastar district, Madhya Pradesh, India. J. Bombay Nat. Hist. Soc. 75: 304–311.

Jaenson, T.G.T., M. J. Gomes, R. C. Barreto dos Santos, V. Petrarca, D. Fortini, J. Evora, and J. Crato. 1994. Control of endophagic *Anopheles* mosquitoes and human malaria in Guinea Bissau, West Africa by permethrin-treated bed nets. Trans. Roy. Soc. Trop. Med. Hyg. 88: 620–624.

Klinkenberg, E., W. Takken, F. Huibers, and Y. T. Touré. 2003. The phenology of malaria mosquitoes in irrigated rice fields in Mali. Acta Trop. 85: 71–82.

Lindsay, S. W., J. R. Armstrong Schellenberg, H. A. Zeiler, R. J. Daly, F. M. Salum, and H. A. Wilkins. 1995. Exposure of Gambian children to *Anopheles gambiae* malaria vectors in an irrigated rice production area. Med. Vet. Entomol. 9: 50–58.

Lindsay, S. W., and R. W. Snow. 1988. The trouble with eaves; house entry by vectors of malaria. Trans. Roy. Soc. Trop. Med. Hyg. 82: 645–646.

Pålsson, K., J. Pinto, V. E. do Rosario, and T.G.T. Jaenson. 1998. The palpal ratio method compared with PCR to distinguish between *Anopheles gambiae s.s.* and *A. melas* from Guinea Bissau, West Africa. Acta Trop. 70: 101–107.

Paskewitz, S. M., and F. M. Collins. 1990. Use of the polymerase chain reaction to identify mosquito species of the *Anopheles gambiae* complex. Med. Vet. Entomol. 4: 367–373.

Reisen, W. K., and F. L. Boreham. 1982. Estimates of malaria vectorial capacity for *Anopheles culicifacies* and *A. stephensi* in rural Punjab province of Pakistan. J. Med. Entomol. 19: 98–103.

SAS Institute. 1996. SAS proprietary software release 6.12. SAS Institute, Cary, NC.

Scott, J. A., W. G. Brogdon, and F. H. Collins. 1993. Identification of single specimens of the *Anopheles gambiae* complex by the polymerase reaction. Am. J. Trop. Med. Hyg. 49: 520–529.

Smith, T., J. D. Charlwood, W. Takken, M. Tanner, and D. J. Spiegelhalter. 1995. Mapping the densities of malaria vectors within a single village. Acta Trop. 59: 1–18.

Trape, J. F., E. Lefebvre-Zante, F. Legros, G. Ndiaye, H. Bouganali, P. Druilhe, and G. Salem. 1992. Vector density gradients and the epidemiology of urban malaria in Dakar, Senegal. Am. J. Trop. Med. Hyg. 47: 181–189.

Trape, J. F., C. Rogier, L. Konate, N. Diagne, H. Bouganali, B. Canque, F. Legros, A. Badji, G. Ndiaye, P. Ndiaye, K. Brahimi, O. Faye, P. Druilhe, and L. Pereira Da Silva. 1994. The Dielmo project: a longitudinal study of natural malaria infection and the mechanisms of protective immunity in a community living in a holoendemic area of Senegal. Am. J. Trop. Med. Hyg. 51: 123–137.

World Health Organization. 1982. Manual on environmental management for mosquito control, with special emphasis on malaria vectors. World Health Organization, Geneva, Switzerland.

Received 20 March 2003; accepted 2 March 2004.